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Dissertation

Designing the airframe and structure

of an

Autogyro for medical transport applications

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Abstract

This project explores the design and development of an unmanned autogiro airframe aimed to transport medical samples which in this case is blood, between hospitals in Sheffield. With an increasing interest to use drones to solve time-critical delivery challenges, particularly in urban settings, the focus here is to create a lightweight, efficient and structurally sound design that can operate safely in the shared airspace.

My role involved designing the airframe and structure of the aircraft. The design had to meet the NHS payload requirements and ensure secure handling of sensitive samples. A push rotor configuration and appropriate materials were chosen to maximize the efficiency and performance. CAD modelling tools like Fusion360 were used to develop the airframe, and the design was validated through structural simulations on ANSYS to check for deformation, stress, strain, vibration, etc.

The process included early-stage modelling, iterative refinements, and eventually building a scaled prototype. Stakeholder feedback was considered to improve the design. The goal was to produce a practical solution that combines safety, performance, and compliance for medical transport in urban environments.

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Nomenclature

μ	Advance ratio	
C_l	Coefficient of Lift	
C_d	Coefficient of drag	
L/D	Lift to Drag Ratio	
E	Young's modulus (Elastic modulus)	Pa or GPa
ρ	Density	kg/m^3 or g/cm^3
ω_{sp}	Short period frequency	rad/s
ζ_{sp}	Short period damping	
$\mu\epsilon$	Microstrain	10^{-6} strain units

1. INTRODUCTION

Urban hospitals often face delays in transporting medical samples or lab materials between buildings or even hospitals. The traditional ground-based methods are sometimes too slow or become unreliable, especially in large campuses or during emergencies. One solution that has come up is the use of unmanned aircraft systems (UAS). In this project I've focused on designing the airframe for an autogyro-based platform that would be capable of meeting this challenge.

Autogyros are a good choice for this kind of task due as they're relatively simple, efficient and stable during flight. Unlike helicopters or fixed wing drone, they can generate lift passively though the main rotor while using a push-propeller at the back for forward thrust. This makes them safer and easier to manage at low-speed airspaces like hospital zones.

The aim of this project is to create a robust yet lightweight airframe that can securely carry a medical payload while meeting structural, aerodynamic, safety requirements. The design process involved CAD modelling, material selection, and structural analysis through ANSYS simulations. The final design will be tested though a scaled prototype, with feedback from NHS-related stakeholders helping to refine the concept.

The aim is to design and validate a robust yet lightweight autogyro airframe capable of securely carrying medical payloads while meeting structural, aerodynamic, and safety requirements.

Objectives

- Develop a lightweight, aerodynamic structure to minimize drag and ensure stability.
- Integrate a push-rotor configuration to optimize thrust and weight distribution.
- Select materials that balance high strength-to-weight ratios with impact resistance.
- Validate the structural design using finite element analysis (FEA) in Ansys.
- Build and test a scaled prototype, refining the design using feedback from NHS stakeholders.

The sections of this report present the methodology, results, and discussion, culminating in conclusions and recommendations for further development.

2. BACKGROUND

An autogyro, or gyroplane, is a rotary-wing aircraft in which the main rotor freely rotates due to autorotation to generate lift. At the same time, a propeller driven by a separate engine provides forward thrust. Unlike a helicopter, an autogyro isn't powered by an engine and instead relies on the upward flow of air through the rotor disc as the aircraft moves forward.

Juan de la Cierva's pioneering work in the 1920s introduced flapping hinge rotor blades, a key innovation that solved the problem of dissymmetry in lift and enabled stable autorotative flight. This simple mechanical solution clearly establishes the operational reliability of an autorgyro.

Autogyros offer several advantages for critical missions in environments packed with constraints, including built-in stall resistance, ability to take off and land on a short runway, and safe descent under engine-out conditions (with the help of autorotation). Their minimal runway requirements make them well-suited for urban missions, and in this case, align closely with the demands of rapid medical logistics.

Early autogyro theories established lift coefficients in the range of 0.5-0.6 during autorotation. Research has progressed over time to high-fidelity flight dynamics modelling, allowing us to capture combined rotor-fixed-wing stability characteristics accurately and address control law design. These durable and modern characteristics make the autogyro a technically viable and an operationally promising option for urgent medical sample transportation in dense urban airspaces.



Figure 2.1: Autogyro Calidus at Shuttleworth Season Premiere 2016 (Airwolfhound, 2016)

3. LITERATURE REVIEW

Knowing the operational advantages there is now a strong foundation as to why autogyros can be used in urgent medical transport. To deliver a structurally strong and efficient, aerodynamically stable and mission capable autogyro having knowledge on airframe structural design, compound rotorcraft aerodynamics, control systems, payload integration, empennage configuration and other multiple disciplines. This section reviews key research contributions in these domains, with the focus on techniques applicable to the design and development of a medical sample delivery UAS.

3.1. Rotor-Airframe Aerodynamic Interaction

The fundamental challenge in designing the structure of an autogyro design lies in managing the aerodynamic interactions. Especially between the unpowered rotor and fixed airframe components. Hamamoto et al. (2022) conducted a comprehensive CFD analysis of the interference experienced by the rotor-wing. The analysis used rFlow3D simulations of a UH-60A rotor above a rectangular wing across advance ratios μ = 0.1-0.7. The results reveal persistent asymmetric loading, with the advancing side experiencing up to 20% lift deficit and altered drag characteristics even at high forward speeds. Root sections show the most severe effects, with the cross-sectional coefficient of lift (C_l) going negative at low advance ratios due to flow reversal.

These findings have direct implications for autogyro structural design, particularly regarding wing root reinforcement and boom-to-fuselage attachment points. The study demonstrated that even at μ = 0.7, the total aircraft lift remains 5% below isolated wing performance. Thereby necessitating conservative structural sizing to accommodate these interaction effects (Hamamoto et al., 2022).

3.2. V-tail versus Twin Boom H-tail Configuration Analysis

Choosing between V-tail and H-tail is critical as having either of the configurations significantly impacts both structural complexity and operational reliability in autogyro when it comes to medical delivery applications. This section focuses on the findings from multiple studies to establish design guidelines for empennage configuration selection.



Figure 3.1: Beechcraft Bonanza in flight near Arlington Airfield (McMahan, 2009)

V-Tail Configurations

The V-tail configuration offers reduced wetted area and potential drag benefits. This done through elimination of horizontal-vertical stabilizer intersection. However, structural analysis reveals critical limitations. Sridharan (2024) documents severe stress concentrations in V-mount structures supporting pusher propellers. FEA analysis using Fusion 360 revealed critical bending points where combined pitch and yaw loads create stress concentrations exceeding material limits. The V-mount design showed susceptibility to fatigue failure under cyclic loading from propeller thrust variations.

O'Brien and Takahashi (2024) demonstrated that V-tail rudder-vators must split elevator and rudder authority between surfaces angled 90 - 110°, reducing effective moment arms for both pitch and yaw control. This geometry demands high deflection angles and increased hinge moments to achieve equivalent control authority compared to dedicated horizontal and vertical surfaces. The integrated nature of V-tail control surfaces complicates field maintenance, particularly critical for medical delivery operations requiring rapid turnaround times.

Twin Boom H-Tail Configurations



Figure 3.2: 434th TCW Bakalar Air Force Base C-119 - Twin-boom aircraft - Wikipedia

Sridharan's (2024) second design iteration demonstrated that transitioning from a V-mount to straight a horizontal boom configuration reduced peak bending stress by 50%. The H-tail

configuration enables direct load transfer from empennage to fuselage without complex bending transitions. This improves the structural efficiency and fatigue life. The study also introduces an innovative split-rudder concept enabling independent component replacement through carbon fiber rod connections, reducing maintenance downtime by >50% compared to integrated V-tail structures.

H-tail boom structures provide superior vibration isolation compared to V-mount configurations. The study documents reduced vibration transmission through ribbed fuselage design and boom-mounted empennage, protecting sensitive medical cargo from propeller-induced disturbances. H-tail configurations enable independent manufacturing and assembly of horizontal and vertical surfaces, reducing production complexity and enabling rapid prototyping using additive manufacturing techniques.

Based on comprehensive analysis, H-tail configurations demonstrate superior suitability for medical delivery autogyros through simplified load paths that reduce stress concentrations, independent pitch/yaw surfaces that maintain control authority during component failures, modular design enabling rapid component replacement, superior vibration isolation protecting medical cargo, and manufacturing flexibility enabling cost-effective prototyping and scaling.

3.3. Advanced Material and Structural Optimization

Composite materials offer substantial advantages for autogyro structures. Particularly in achieving the weight efficiency critical for medical delivery missions. Kassimov et al. (2025) demonstrated that through CFD analysis and experimental testing, carbon fiber composites (ρ=1.55 g/cm³, E =230 GPa) help in providing 40% structural weight reduction. This contributes to a 35% flight duration increase compared to aluminum structures. Their study shows carbon composites achieve Cl/Cd=5.4 at low speeds. This reduces the drag coefficient from 0.025 to 0.018 and vibration amplitude from 0.7mm to 0.4mm.

The integration of CAD/CAE/CFD methodologies demonstrated by Kassimov et al. (2025) provides a validated approach for autogyro structural optimization. The comparisons show that carbon composites halve the drag coefficient while boosting lift coefficient by 33% at 30 m/s airspeed, directly applicable to autogyro cruise performance enhancement.

Elelwi, Botez and Dao (2021) apply topology optimization using SIMP methodology to

achieve over 50% structural mass reduction in morphing wing designs. Their method of combining CFD-derived load distributions with OptiStruct structural optimization demonstrates how aerodynamic loading can inform material placement for maximum efficiency. For autogyro applications, this approach allows us to optimize the rotor mast pylons, boom structures, and empennage components under realistic flight loads.

Sridharan (2024) demonstrated that the application of carbon fiber composites in UAV structural components helps in achieving high strength-to-weight ratios while enabling 3D printing of complex geometries. The study's comparison of PLA versus premium plastic materials (polycarbonate, ABS, carbon-fiber-reinforced polymers) shows substantial improvements in durability, heat resistance, and vibration dampening essential for medical delivery operations.

3.4. Adaptive and Morphing Structures

Shape-adaptive technologies offer potential performance benefits for autogyro medical delivery missions. İnsuyu (2010) provides the most comprehensive study of mission-adaptive wings, developing a 2m-span UAV wing with ±12% chord camber variation over 33% span. Using ANSYS Fluent CFD analysis of 191 camber variants, the study achieves 25% L/D improvement at optimum settings while maintaining structural integrity through modal analysis (first bending mode at 22.4 Hz) and flutter analysis (divergence at ~200 m/s). The integrated aero-structural design methodology, validated through TAI prototype manufacturing, establishes a framework directly applicable to autogyro control surface optimization.

3.5. Multidisciplinary Design Optimization (MDO) and AI Integration

Söbester and Keane (2006) demonstrated the practical implementation of a CAD-driven MDO using CATIA V5 geometry coupled with VSAERO aerodynamics and ABAQUS structural analysis. Their parametric design workflow which incorporated the Breguet range equations, crash energy criteria, and constraint analysis was able to achieve optimal washout and sweep configurations (5°/20°) for a blended-wing UAV while maintaining structural deflections under 3 mm at 3.8g loading. This methodology directly applies to autogyro empennage and boom optimization.

The study's factorial design of experiments approach provides a systematic methodology for

autogyro configuration optimization, demonstrating how CAD-driven parametric models enable rapid exploration of design space while maintaining structural feasibility constraints (Söbester and Keane, 2006).

Karali, Inalhan and Tsourdos (2024) demonstrated an AI-driven multidisciplinary conceptual design using neural network surrogates which was trained on OpenVSP geometry, VSAERO aerodynamics, and structural mass predictions. Its multi-objective genetic algorithm rapidly explored the design space for wing aspect ratio, taper, and sweep. While at the same time optimizing L/D, weight, and radar cross-section. This methodology enables rapid autogyro configuration studies integrating rotor-airframe interactions with structural requirements. Latin-hypercube design of experiments generates 1,000 geometry-flow cases for neural network training. Thereby providing a framework scalable to autogyro-specific design parameters including rotor mast placement, boom geometry, and empennage sizing.

3.6. Payload Integration and Medical Delivery Systems

Developing the structure for medical delivery drones imposes unique structural requirements such as vibration isolation, thermal stability, and crash protection. Saponi et al. (2022) investigated payload solutions for delivering medium and small items weighing up to 5 kilograms. This was done using a medium-sized UAV with a maximum takeoff weight (MTOW) of less than 25 kilograms. The study focused on payload solutions integrated within drone fuselage, medical delivery compatibility, and user-centered design factors. The prototype developed industrialized package design and uses high-performance double-wall cardboard balancing strength and weight for UAV delivery applications.

Sadiq, Iqbal and Shah (2016) addressed servo-actuated payload carry and drop mechanisms for unmanned helicopters in disaster relief operations. They conducted virtual simulations to test the design using SolidWorks. This was able to demonstrate payload drop mechanisms capable of precise delivery in inaccessible terrain, directly applicable to medical supply delivery in emergency scenarios.

Kumar, Agrahari and Bucha (2024) specifically address UAV design for medical relief material delivery, establishing performance requirements and structural considerations for hospital-to-hospital transport missions. Their analysis emphasizes on lightweight yet robust airframes capable of protecting sensitive medical cargo during transport and emergency

scenarios.

Sridharan (2024) addresses UAV design specifically for medical blood sample delivery between Sheffield Children's NHS Foundation Trust and Royal Hallamshire Hospital, establishing performance requirements including reduced vibration and shortened journey times. The payload design incorporates a sliding mechanism with an automatic locking system and shock-absorbers. It also comes with retention mechanisms to prevent cargo movement during flight maneuvers. This approach to medical cargo protection provides a framework applicable to autogyro payload bay design which can then be carried forward while designing the structure of the autogyro .

3.7. Structural Analysis and Validation Methods

Jayakumar et al. (2024) examined the structural-aerodynamic coupling in compact ducted UAVs using ANSYS Workbench FSI analysis. The material comparison study shows that titanium alloy achieves the lowest deformation (0.00012 mm) under aerodynamic loads, though weight penalties must be balanced against performance requirements. The study's integration of CFD pressure distributions with structural analysis provides a methodology applicable to autogyro fuselage and nacelle design.

3.8. Hybrid UAV Design and Conceptual Methodologies

Varsha and Somashekar (2018) presented a high-performance UAV conceptual design methodology incorporating constraint diagrams, stability analysis, and structural sizing for a 550 kg platform. Their systematic approach from constraint analysis through to powerplant selection and propeller design established benchmarks.

3.9. Stability and Control Requirements

O'Brien and Takahashi (2024) establish stability screening criteria for high-speed UAVs, emphasizing adequate static margins$\left(-\frac{dC_m}{dC_l}\right)$, , short-period frequency ($\omega_{sp} > 1$ rad/s), and damping ($\zeta_{sp} > 0.35$) for Category-I performance. Their analysis of lateral control departure parameters and inertial coupling ratios provides guidance for empennage sizing to ensure controllability under emergency conditions. The study's application of MIL-STD-8785C criteria to slender high-speed configurations demonstrates the critical importance of control surface authority and structural stiffness for maintaining flying qualities under rotor-induced

loads.

3.10. Synthesis and Future Direction

The literature reveals convergence toward integrated design methodologies combining CFD-informed load analysis, topology optimization, advanced composites, and AI-driven configuration studies. For medical delivery autogyros, key structural design considerations include: managing rotor-airframe interaction loads through reinforced attachment points accounting for 5-20% load increases due to rotor wake interference; exploiting composite materials for 30-40% weight reduction and 35% flight duration increases; implementing H-tail configurations for simplified load paths, enhanced control authority, and modular maintenance reducing downtime by >50%; integrating morphing technologies providing up to 25% L/D improvements; and developing specialized payload integration systems with vibration isolation, thermal protection, and crash survivability.

Future research should focus on developing autogyro-specific load cases incorporating autorotation dynamics, validating composite structures under combined rotor and gust loads, establishing certification standards for medical payload protection during emergency landings, and integrating AI-driven optimization with high-fidelity rotor-airframe interaction modeling for next-generation autogyro structural design.

4. METHODOLOGY

Designing an advanced autogyro UAV for medical logistics in urban environments requires a careful, iterative methodology that integrates the airframe and offers control, system level reliability, decent aerodynamic performance and most importantly payload protection. The approach taken for this project followed aerospace design cycles, but aspects of the workflow were slightly unconventional such as beginning from the outer frame before the chassis of the autogyro. This influenced the design outcomes and reflections which will be presented later in this report. This chapter details the methodology step by step, closely linked to the project aims of producing a lightweight, robust, and safe airframe capable of carrying the medical payloads in constrained urban environments.

4.1. Objectives, Requirements, and Conceptualization

The first step involved was to clarify the operational requirements and constraints. The Payload integration for medical samples set the baseline. The design needed to house the avionics including parts of the power and lift mechanism which were carried over from last year such as the power module, flight computer, T-Motor AM480 – the rear motor for the push propeller, etc. Due to this the maximum take-off weight was set to 9kgs. This meant that the complete aircraft including the payload could only weigh a maximum of 9kgs. This target framed all structural decisions even though later prototypes exceeded this threshold.

Initial design inspirations were drawn from existing UAV frames, focusing on the twin-boom layouts and circular frame concepts. Early sketches aimed to integrate payload bays while balancing the rear rotor's clearance, tail configuration and the overall packaging.



Figure 4.1: Hanseatic S360 fixed-wing drone (Hanseatic Aviation Solutions GmbH, 2021)

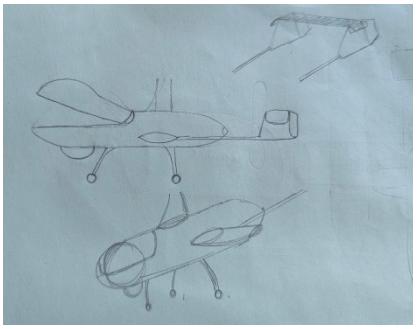


Figure 4.2: Initial sketches of the possible Design approaches

The design process began with the outer frame and then progressed to the chassis. Although like mentioned before the process was unconventional but allowed rapid visualization of the

aircraft's proportions and highlighted spatial allowances for avionics. However, this also created excess empty volume that could have been avoided with a chassis-first approach which would have in turn helped in reducing the overall mass.

During the later stages of the design process an "Iron Bird" was constructed using 6mm MDF and Birch Plywood sheets which were laser cut using AutoCAD and Lightburn. The primary purpose of building one was to check fit and see if the avionics and the legacy components integrated well rather than assessing the load bearing behavior. Another point to keep in mind while designing the overall structure of the autogyro, was to set its minimum and maximum dimensions. As mentioned earlier, the rotor blades were selected early on and were of a total length of 1.2m. To work alongside these constraints a minimum and a maximum needed to be stablished primarily for the length and breadth of the fuselage. Unfortunately, only during the prototyping stage was the size of the fuselage found to be way too big for the rotors chosen. Nevertheless, this simplified model of an Iron confirmed that the placement of the propulsion system, power electronics, and the control units could be achieved without interference but with a lot of empty space and that the overall structure needed to be scaled down to meet the dimensions of the rotor blades.

4.2. CAD, Preliminary Structural Definition and Material Selection

Going by the conventional process, CAD models are to be developed following Iron Bird validation. This stage involved designing the main frame, chassis, and empennage geometry while focusing on the weight distribution and structural connectivity. Designing it on Fusion360 provided parametric flexibility, allowing repeated refinements.

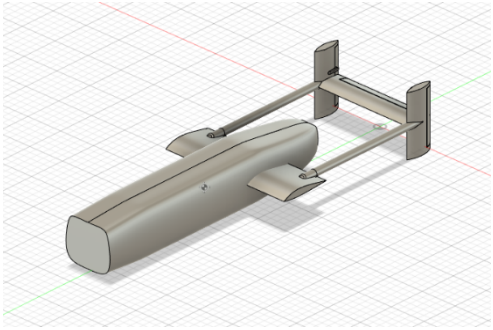


Figure 4.3: First versions of the CAD models developed. The tail was taken from the CAD model of the ABABIL-3 drone

The empennage configuration was a key decision point. Based on comparative research and Sridharan Sakthivel (2024), the H-tail provided better yaw authority in the push-propeller slipstream, simplifies actuation by avoiding control-surface mixing, and reduced

maintenance complexity. Though slightly heavier in wetted areas, the structural efficiency in carbon fiber reinforced polymer (CFRP) laminates outweighed potential drawbacks.

Material selection was a pivotal phase in weight reduction alongside manufacturability and cost feasibility. Three materials were evaluated:

- **Carbon fiber reinforced polymer (CFRP):** High stiffness-to-weight ratio and anisotropic tailoring capabilities. FEA helped confirm that a full CFRP structure would withstand flight loads with minimal deflection. However, cost and fabrication limitations made full-scale CFRP unrealistic for this stage.
- **Aluminum:** Used for the circular frames around the main chassis. Provided rigidity at high-load interfaces. While reliable and slightly more cost effective compared to CFRP, it adds significant mass relative to the other components.
- **Glass fiber reinforced polymer (GFRP):** Applied to non-critical surface such as the holder to keep the avionics in place and the payload holder under the chassis. It offers low manufacturing costs and intrinsic damping but greater deflection in structural members.

While a hybrid strategy (CFRP for load paths, aluminum for junctions, GFRP for panels)was considered during design, all FEA simulations were carried out assuming the full airframe was made from CFRP. This was necessary to provide consistent numerical results to evaluate upper limit case for stiffness and strength. In practice, however CFRP was not applied due to cost and fabrication limitations.

4.3. Finite Element Analysis and Structural Validation

FEA was chosen in case as the primary tool for analysis because traditional analytical methods (beam or plate theory) would not be able to capture the combined influence of gravity and the other applied loads acting on the multi-composite structure.

Finite Element Analysis

Finite Element Analysis (FEA) is a computational method used to approximate the response of complex structures under loading. The geometry is split or discretized into finite elements, each connected at nodes, with equilibrium relationships assembled into a global stiffness matrix:

$$[K]\{u\} = \{F\}$$

where $[K]$ is the global stiffness matrix, $\{u\}$ the nodal displacement vector and $\{F\}$ the applied force vector. Once displacements are solved, strains are obtained through geometric relations, and stresses are computed using Hooke's Law:

$$\{\sigma\} = [D]\{\epsilon\}$$

where $[D]$ is the material property matrix and $\{\epsilon\}$ the strain vector. This framework makes FEA particularly suited to aerospace applications, where structures are geometrically complex, loads are multi-directional, and materials such as CFRP are orthotropic. By applying FEA, the structural integrity of the autogyro airframe could be assessed virtually, guiding design decisions before committing to manufacturing.

Simulation Setup

The baseline geometry was designed on Fusion360 as mentioned before and then imported into ANSYS as a .STEP file. Pre-processing involved removing unnecessary small features to improve mesh quality which was done through Design Modeler and Discovery. The Components were classified into:

- **Beam-like members:** CFRP tubular structures, modelled with beam elements to capture bending and torsional stiffness with reduced mesh cost.
- **Thin-walled panels:** fuselage skins and empennage surfaces, represented with shell elements to capture in-plane stiffness and out-of-plane bending.

This strategy ensures accuracy in load paths while maintaining manageable model size.

Material properties

The primary structural material was Epoxy Carbon UD (E-230GPa). An orthotropic material definition was used with a Longitudinal modulus of ~230GP and transverse moduli and shear constants which were representative values from epoxy-carbon composites. This definition avoided isotropic simplifications, and provided realistic stiffness in the fibre and matrix directions

Meshing Strategy

A multi-resolution mesh strategy was adopted where:

- **Global mesh:** Baseline element size was chosen for a balance between model accuracy and computational load.

- **Local refinement:** applied at high-stress regions such as the boom-fuselage junctions, empennage connections, etc.

Mesh convergence helped in verifying local stresses were mesh independent.

Loads and Boundary Conditions

Multiple Loads were designed to reflect both in-flight and support conditions.

- **Gravity Loads:** applied globally to represent the weight itself.
- **Lift forces:** distributed across the wing supports and rotor mountings to simulate aerodynamic lift.
- **Thrust forces:** applied as concentrated loads at the pusher-propeller mount, aligned with the longitudinal axis.
- **Payload weight:** represented as point forces at the payload bay.

Boundary conditions:

- **Fixed supports:** applied at the ground test attachment points
- **Elastic supports:** modelled around spring foundations and are used to simulate realistic joint movement while avoiding be overly constrained.

Load Scaling for Operational Representatives

Initial runs showed gravity dominance over aerodynamic loads. To ensure stresses matched the real in-flight conditions, load magnitudes were scaled with the help of realistic safety-factors, enhancing the identification of potential weak points under representative stress conditions. The simulation was all carried out as static structural cases with geometric nonlinearity activated to account for large deflections. The main cases include:

1. Gravity-only (baseline check)
2. Gravity + payload (ground condition)
3. Combined gravity, lift, and thrust (flight condition).

As the first two cases didn't yield results that could be analyzed in depth, only the last case was simulated.

4.4. Payload Integration and Protection

Given the medical logistics focus, payload integration was prioritized. The Iron Bird mock-up validated the physical space and fit of the avionics and payload bay components. To limit vibration transmission, the non-critical surfaces were constructed in GFRP, which provided

inherent damping. Active isolation and thermal management methods (e.g., phase-change materials) were considered but not implemented within scope. Nevertheless, the CAD design allowed for future integration. Payload release mechanisms were reviewed through literature, but physical testing was deferred. The methodology thus ensured the payload was structurally integrated leaving subsystem refinement for subsequent development.

4.5. Prototype Development and Iterative Refinement

A scaled prototype was fabricated to test manufacturability and assembly. As seen in Fig. 4.4 and Fig. 4.5, this prototype combined the Iron Bird chassis and avionics board with circular frames from the CAD model. Construction materials included 6mm birch plywood and MDF which were laser cut at iForge.



Figure 4.4: The mainframe laser cut from Birch ply and MDF at the iForge

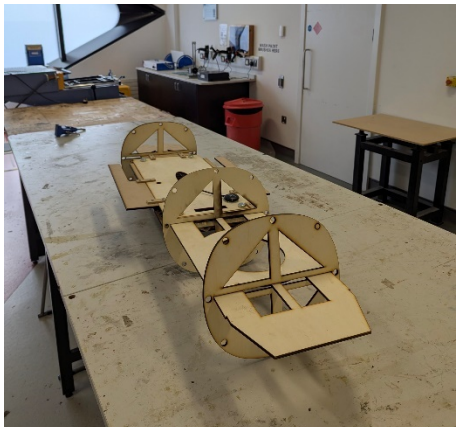


Figure 4.5: Mainframe fitted with laser cut circular supports

The prototype verified integration of avionics and confirmed assembly processes. It revealed the excess empty space from the outer-frame-design sequence, sagging at the front bit of the chassis and other structural flaws. While it did not meet the weight threshold, it provided critical validation of spatial layout, manufacturability, and assembly feasibility.

The unconventional choice of beginning with the outer frame helped in visualizing how the autogyro could look but it created inefficiencies in the mass distribution. Material selection proved vital to performance, while a hybrid CFRP-aluminum-GFRP approach delivering structural safety albeit at a higher mass than target and would not be economical.

Overall, the following the right methodology would ensure that the design could be structurally viable, manufacturable at prototype scale, and integrated with avionics and payload systems. The findings set a foundation for future teams and their design iterations to focus on weight reduction and aerodynamic refinement while providing a robust design that can protect the payload.

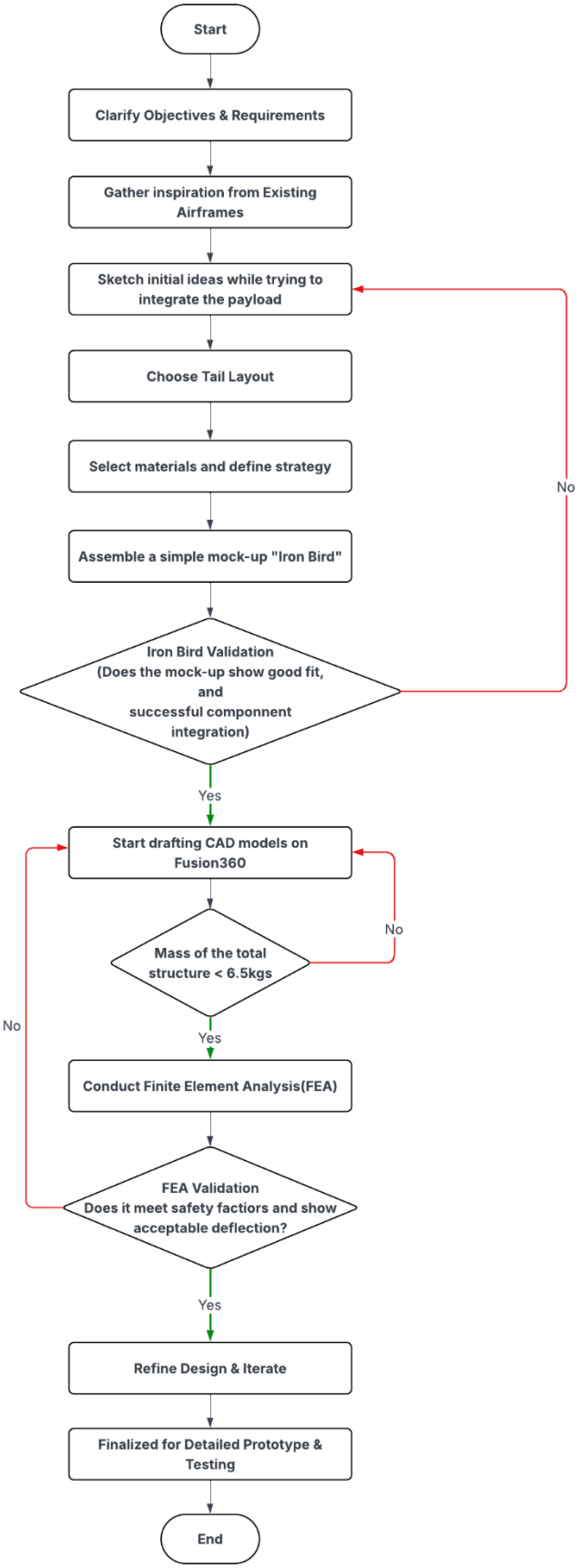


Figure 4.6: A flowchart explaining the conventional methodology

5. RESULTS AND DISCUSSION

5.1. Final Design Configuration

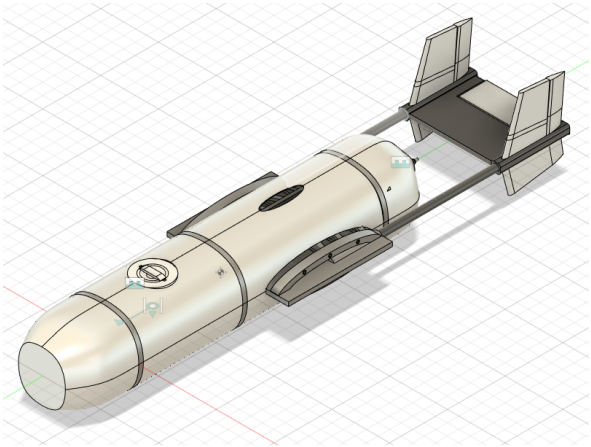
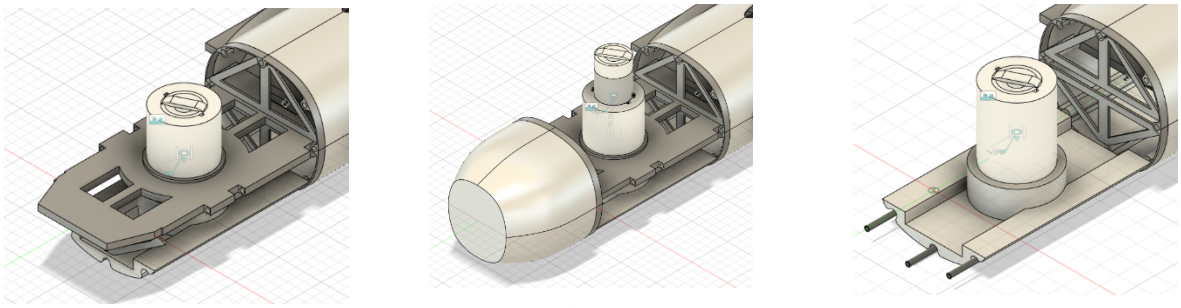


Figure 5.1: Final CAD design incorporating the payload and twin-boom H Tail

The final design put forward was a complete chassis and support package, developed to balance payload integration with structural stiffness, and serviceability. The design incorporated the following:

Payload Integration: At the front of the chassis, and opening was created for the payload bay. The payload would sit on a dedicated sub-structure beneath the chassis helping it locking securely in place. This ensured stability during the operation and safe transportation of medical samples.

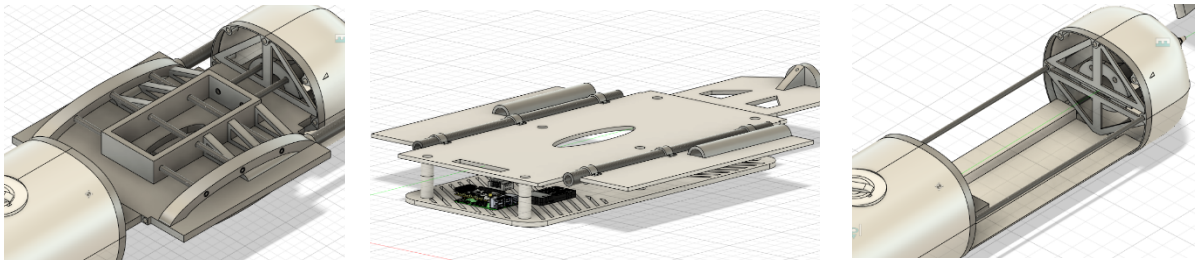


(a) *(b)* *(c)*

Figure 5.2: (a)Payload seated in the cut-out (b) twist and lock mechanism while seated in the fuselage (c) the sub-structure beneath the chassis.

Central Section and Wing Box: The mid-section was designed to withstand the combined demands of supporting the mast, the tail, and protect the electronic systems. A wing-box style arrangement as seen in Fig. 5.3(a) with carbon fiber (CF) tubes that connected the left and right-wing sides improves the stiffness and distributes the loads effectively. While creating the iron bird, a different avionics board Fig. 5.3(b) was designed but wasn't incorporated into the final design. Rather a holder similar to the one for the payload was made which slides in easily as shown in Fig. 5.3(c). This is important as the tail booms which

connect the tail, and the fuselage are connected to this section. To house the avionics a second level was introduced below the main chassis. This allows easy access and facilitates quick replacement of electronic components.



(a) (b) (c)

Figure 5.3: (a) shows a central section for the mast along with the wing box (b) the avionics board below the mainframe in the iron bird (c) the avionics board under the wing box in the final design

Tail and Propulsion Support: At the rear, a structural support, shown in Fig. 5.4, was incorporated to hold the pusher motor. The cutouts in the chassis help in facilitating easy wiring.

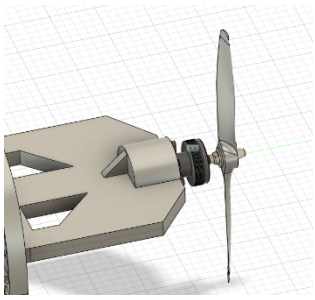


Figure 5.4: Support to hold the pusher motor

Chassis Beam and Strut System: Throughout the chassis, CF tubes were arranged in a beam-and-strut framework, like an aircraft fuselage. This network of tubular members acting as the primary load path, resisting bending and torsional loads while maintaining lightweight efficiency.

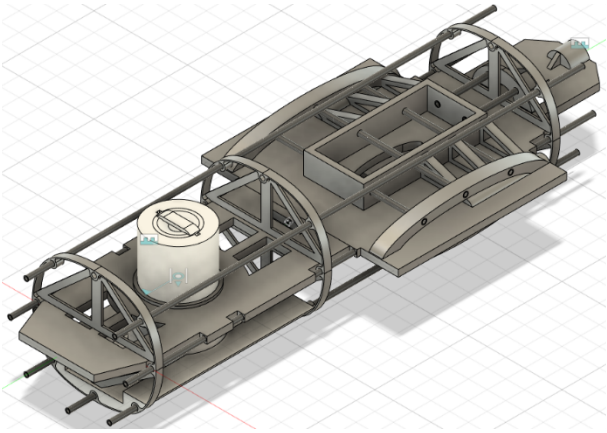


Figure 5.5: Chassis beam and strut system

Outer Skin: As seen in Fig. 5.1, the external skin enclosed the chassis, with cut-outs integrated to allow CF tubes to pass through. Unlike conventional approaches where the skin

is usually the final layer to be designed, the skin was considered from the outset of the design. The dimensions of the final design are presented in Table 5.1 below:

Dimension	Value
Overall length (nose to tail)	1900 mm
Maximum external width	525.72 mm
Maximum external height	300 mm (fuselage), 419.9 mm (tail)
Boom-attachment spacing	400 mm
Payload bay internal length	154.82 mm
Payload bay internal width	154.82 mm
Payload bay internal height	276.04 mm

Table 5.1

5.2.Structural Simulation Results

Finite Element Analysis (FEA) was performed on four structural configurations; each modelled as a complete CFRP body:

- **Structure A - Initial Mainframe Chassis:** the baseline full-span chassis
- **Structure B - A shortened mainframe chassis**: the initial chassis was reduced from 1.4m to 1.1m and the overall weight was reduced by 2kgs from 14kgs.
- **Structure C - Reinforced Mainframe**: the mainframe was reinforced with circular, vertical and tube supports to provide additional stiffness.
- **Structure D - Full body with Tail booms and H-tail**: this was the final configuration with the tail assembly excluding the outer skin of the autogyro.

Each body was subjected to two load cases:

- **Case 1:** Lift ≈ 80-100N, Thrust 40N

This was to simulate the lift and forward thrust produced by the power and lift set-up from last where the motor and rotor chosen were specifically for a maximum take-off weight of 9kg.

- **Case 2:** Lift ≈ 550-600N, Thrust 60-80N

As the final design exceeded the 9kg weight constraint, the thrust and lift generated by the selected propulsion system were too small. To address this, the lift and thrust were increased to ensure the structures were evaluated under realistic operating conditions. Results for each structure are presented in terms of total deformation, equivalent (von Mises) stress, and equivalent elastic strain.
It is important to mention that all values were taken from a true scale of 1.

5.3. Structure A – Initial Mainframe Chassis

Deformation Response

- 0.97 mm under Case 1, meeting the 1mm serviceability limit.
- 4.95 mm under Case 2, nearly five times the limit. Maximum deflection occurred at the mid-span of the chassis tube.

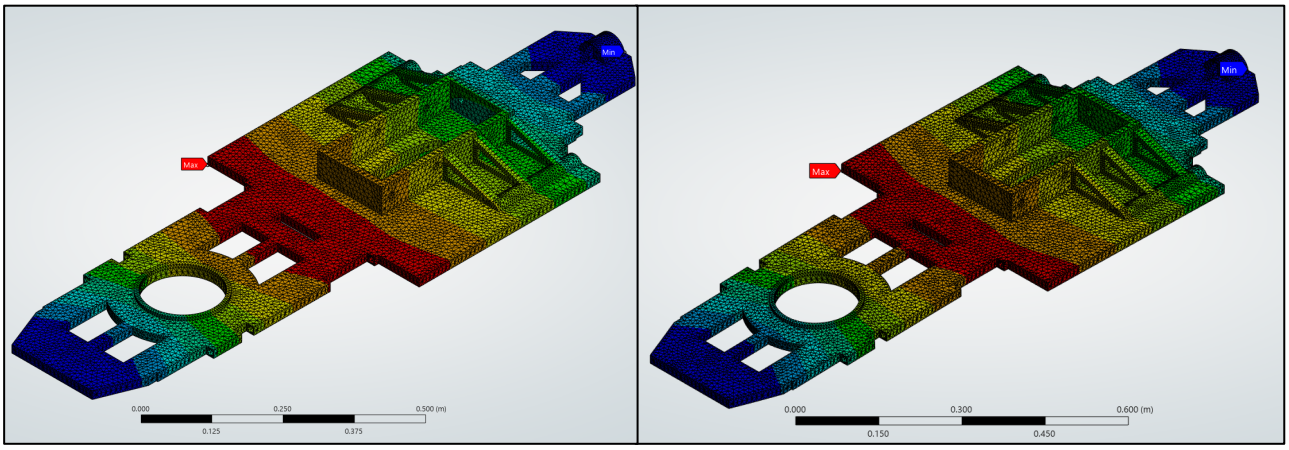


Figure 5.6: Total Deformation for Case 1(L) and Case 2(R) with a scale factor of 100

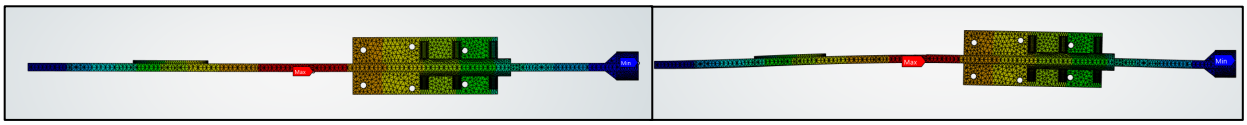


Figure 5.7: Side-view of total deformation for Case 1(L) and Case 2(R) with a scale factor of 100

Stress Response

- Peak stress = 15.6 MPa (Case 1) and 21.9 MPa (Case 2)
- Both values <1% of CFRP tensile strength. Stresses localized at tube-plate junctions.

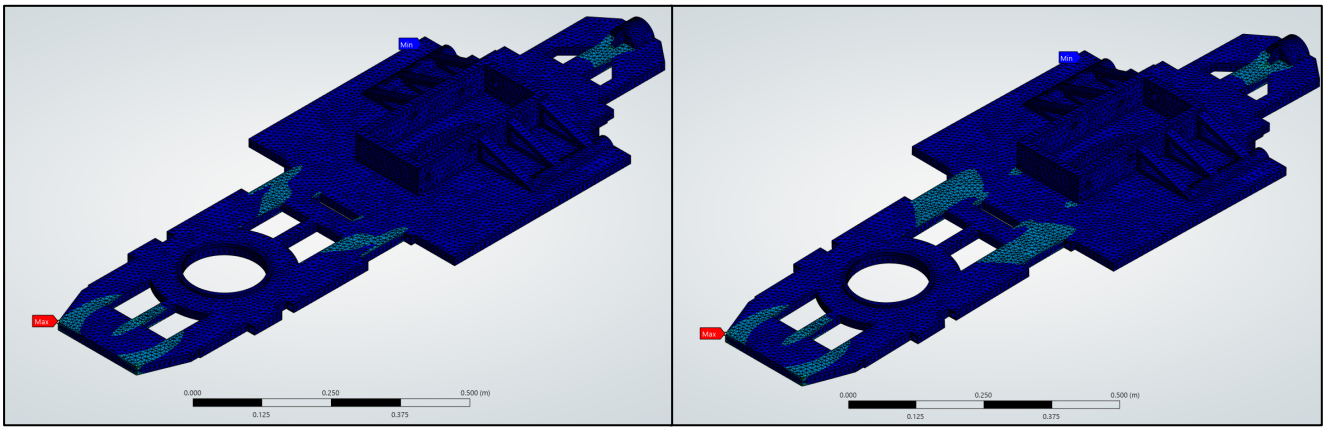


Figure 5.8: Equivalent von-mises stress for Case 1(L) and Case 2(R) with a scale factor of 100

Strain Response

- Equivalent strain = 1358$\mu\epsilon$ (~12% of ultimate strain)

- Strains remained well below failure thresholds, but localized strain concentrations were evident.

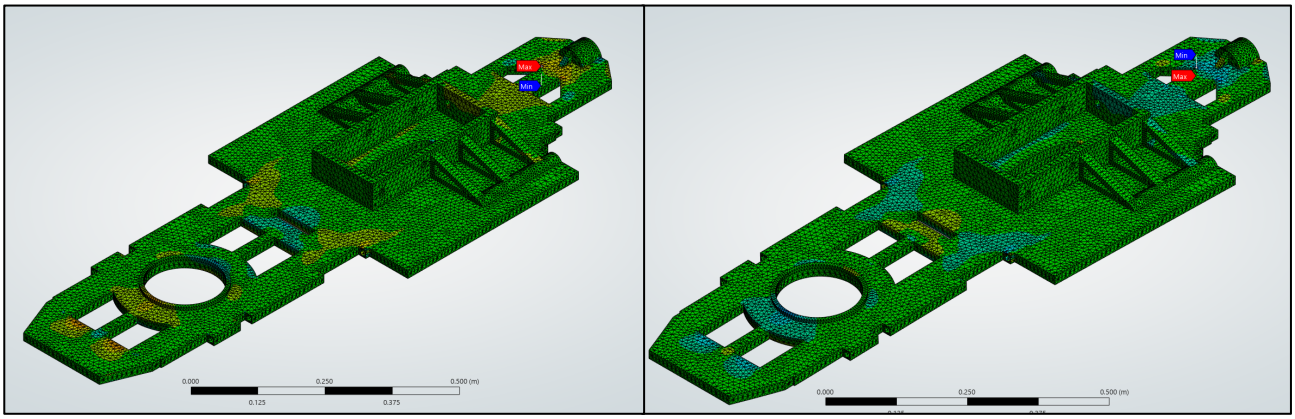


Figure 5.9: Equivalent Strain for Case 1(L) and Case 2(R) with a scale factor of 100

5.4. Structure B – Shortened Mainframe Chassis

Deformation Response

- 0.12mm (Case 1) and 0.97mm (Case 2)
- Both values within 1mm limit, confirming high stiffness.

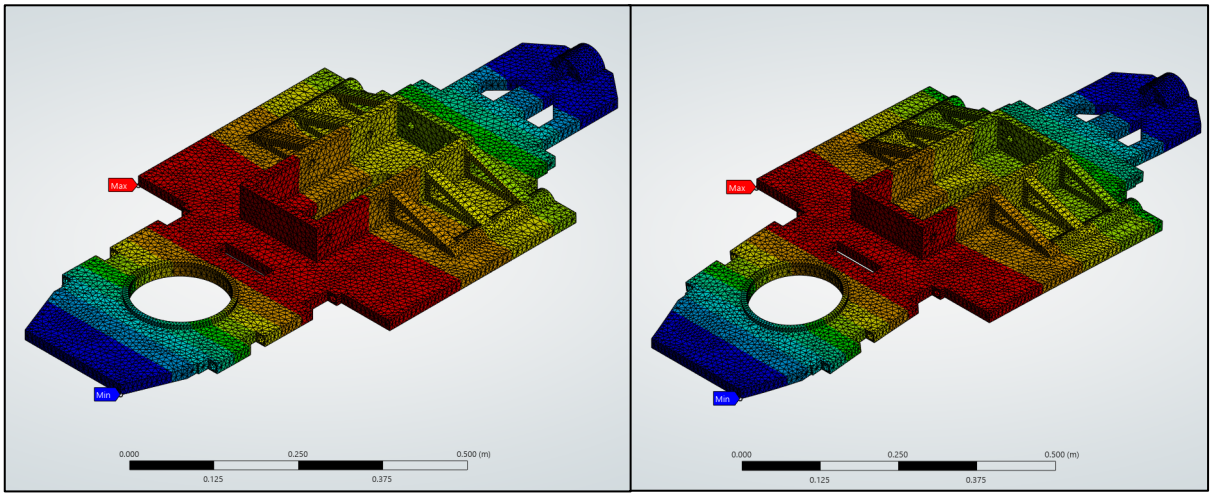


Figure 5.10: Total Deformation for Case 1(L) and Case 2(R) with a scale factor of 31

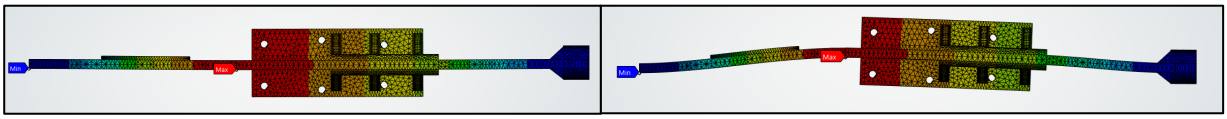


Figure 5.11: Side-view of total deformation for Case 1(L) and Case 2(R) with a scale factor of 31

Stress Response

- Peak stress = 0.59 MPa (Case 1) and 5.85 MPa (Case 2)
- Values significantly below CFRP limits, distributed more uniformly compared to Structure A.

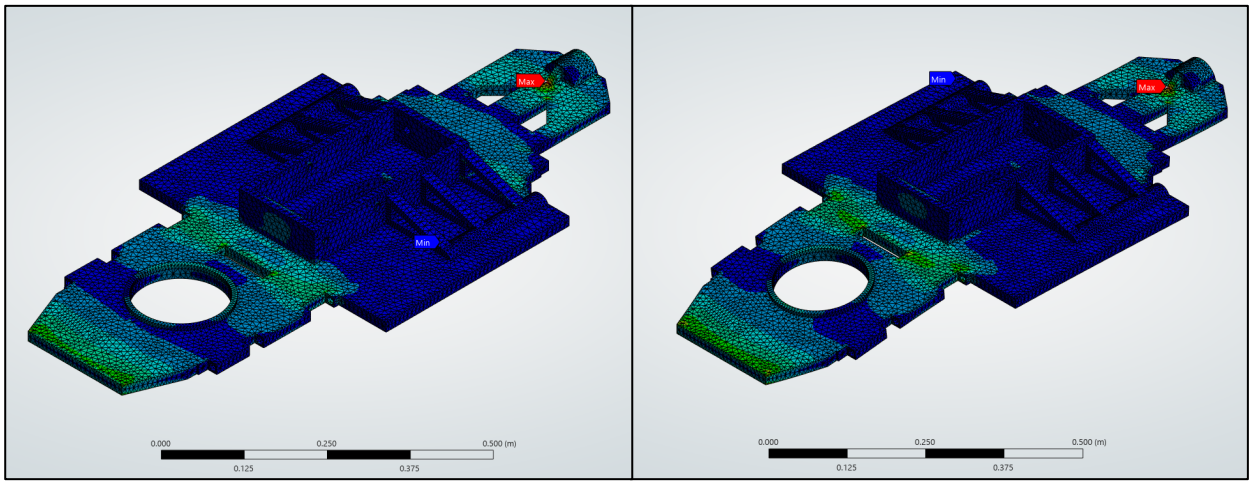


Figure 5.12: Equivalent Von-mises stress for Case 1(L) and Case 2(R) with a scale factor of 31

Strain Response

- Peak strain = 1358$\mu\epsilon$ (~12% of ultimate strain)
- Demonstrates comfortable margins and validates geometry-driven stiffness gains

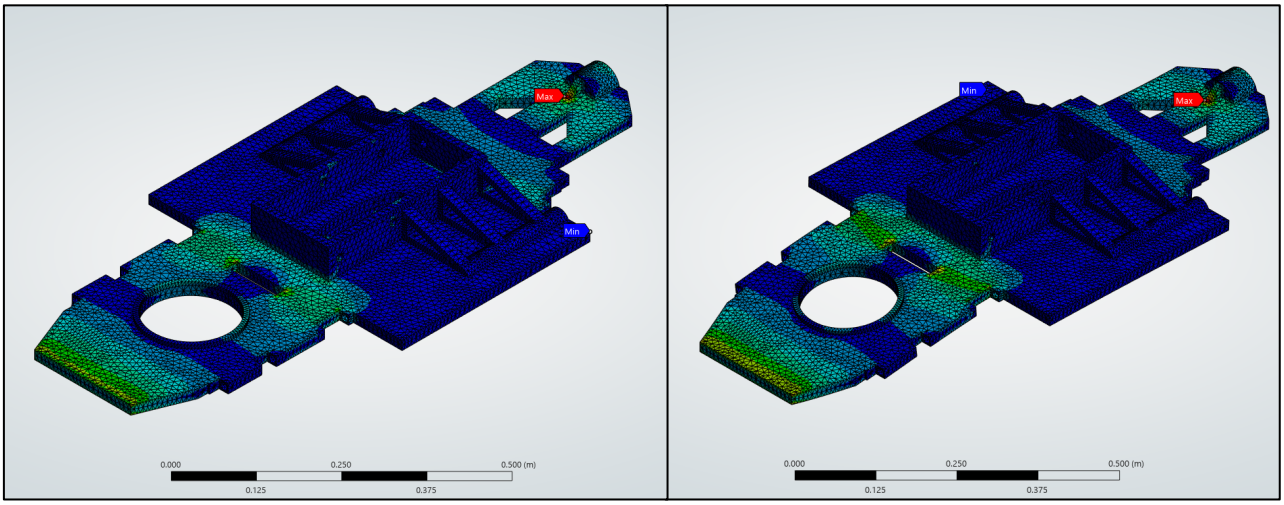


Figure 5.13: Equivalent Stress for Case 1(L) and Case 2(R) with a scale factor of 31

5.5. Structure C – Reinforced Mainframe

Deformation Response

- 0.12mm (Case 1) and 1.56mm (Case 2)
- Exceeds serviceability under Case 2. Reinforcement improved stiffness but did not fully control large-span flexibility

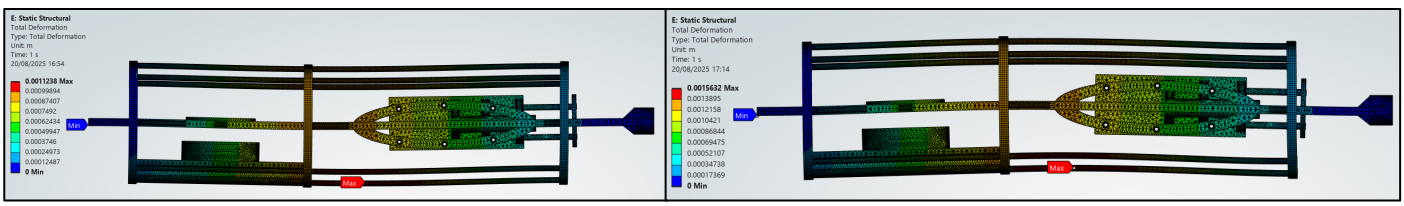


Figure 5.14: Side-view of total deformation for Case 1(L) and Case 2(R) with a scale factor of 10

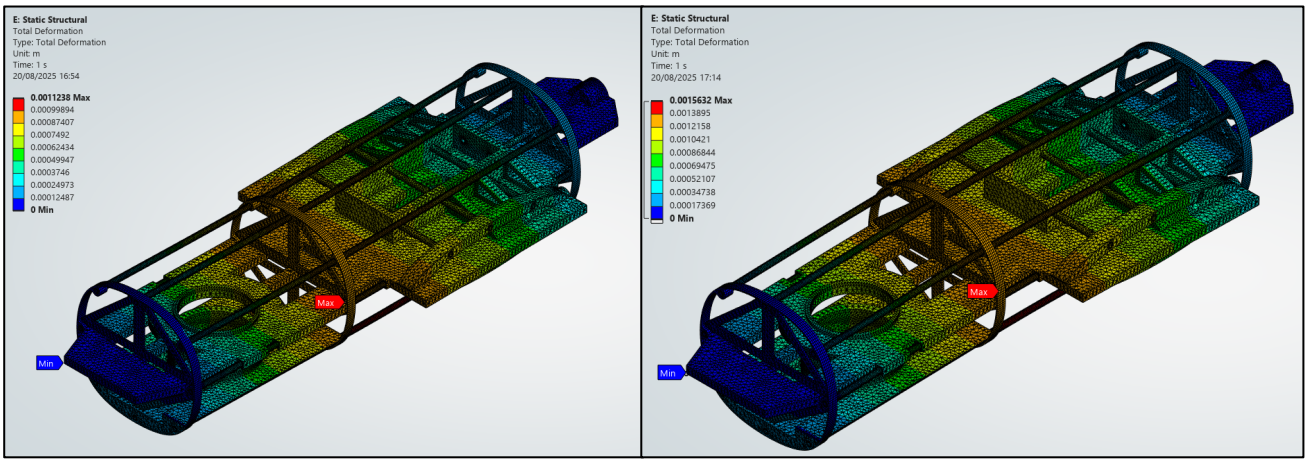


Figure 5.15: Total Deformation for Case 1(L) and Case 2(R) with a scale factor of 10

Stress Response

- 5.85 MPa (Case 1) and 21.9 MPa (Case 2)
- Stress concentrations shifted to reinforcement joints, raising concerns of stress risers

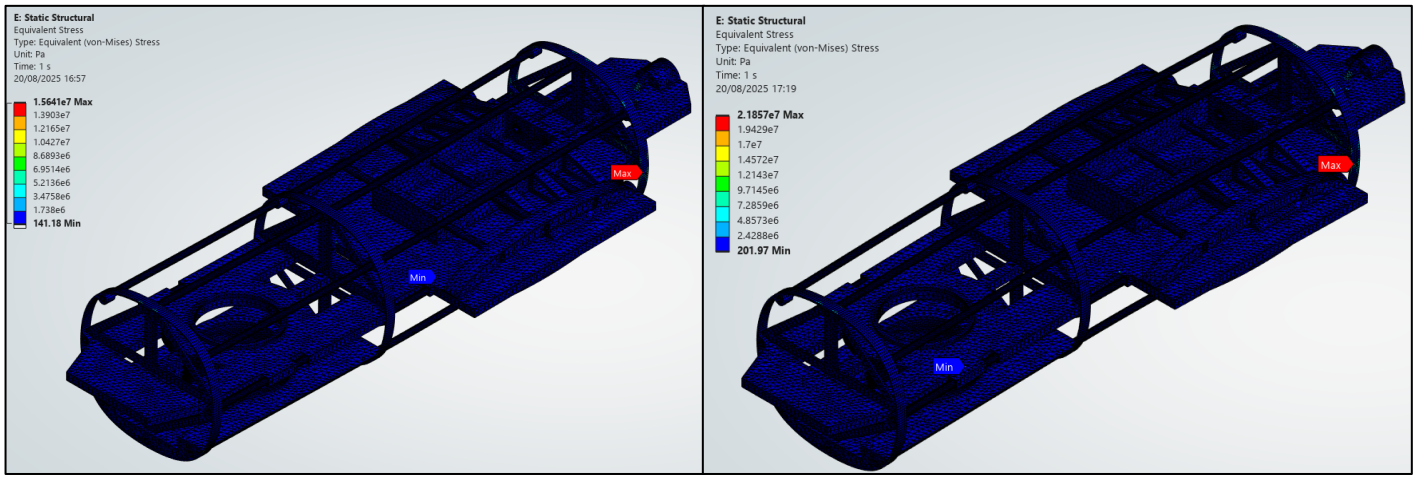


Figure 5.16: Equivalent Von-mises stress for Case 1(L) and Case 2(R) with a scale factor of 10

Strain Response

- 91$\mu\epsilon$ (Case 1) and 1,888$\mu\epsilon$ (Case 2), ~17% of ultimate strain.
- Approaches ply-level damage thresholds, requiring laminate failure analysis

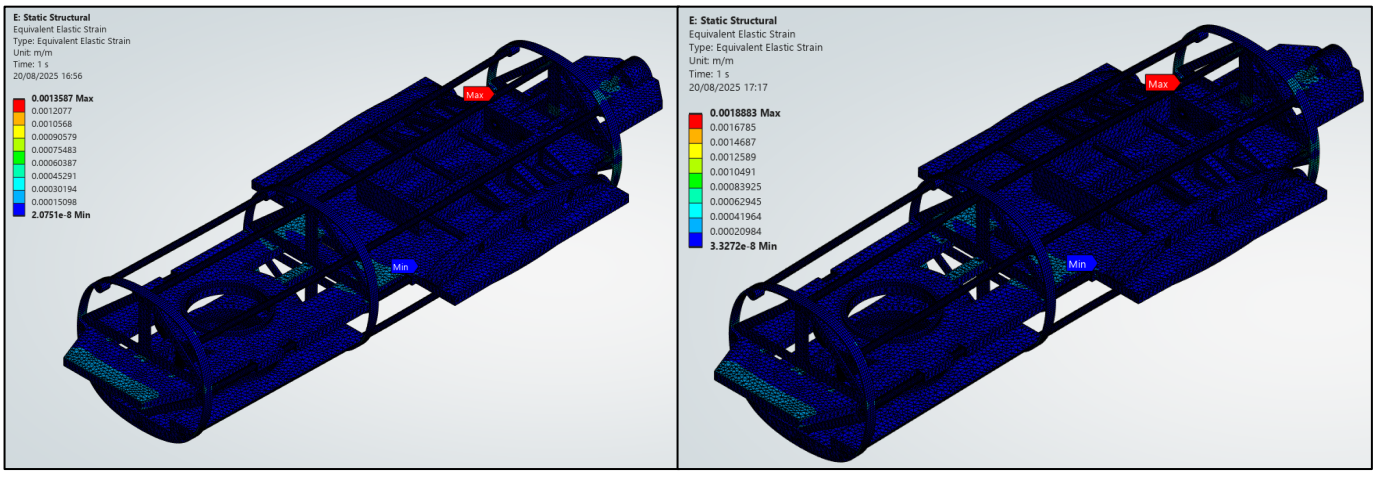


Figure 5.17: Equivalent strain for Case 1(L) and Case 2(R) with a scale factor of 10

5.6. Structure D – Reinforced Mainframe

Deformation Response

- 0.02mm (Case 1) and 0.16mm (Case 2)
- Deflections were negligible and far below the 1mm serviceability threshold.

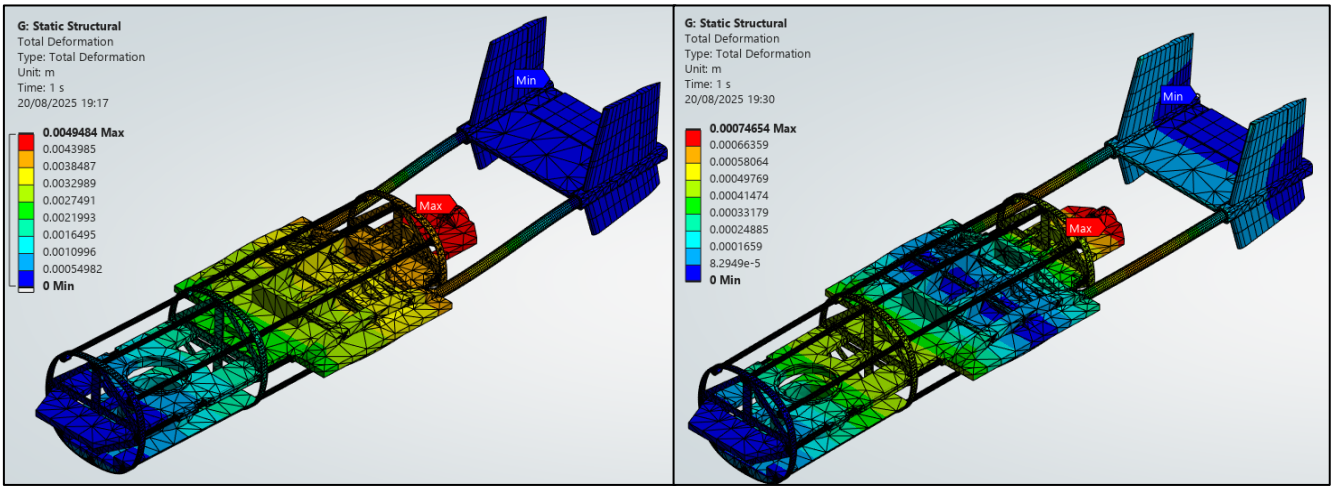


Figure 5.18: Total Deformation for Case 1(L) and Case 2(R) with a scale factor of 10

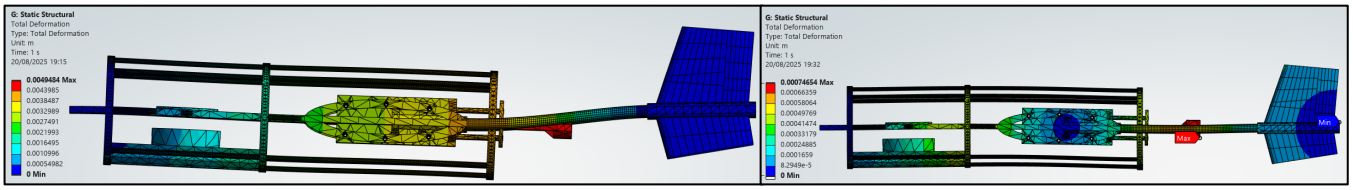


Figure 5.19: Side-view of total deformation for Case 1(L) and Case 2(R) with a scale factor of 10

Stress Response

- 5.15 MPa (Case 1) and 30.4 MPa (Case 2)
- Localized stresses around propeller mount regions

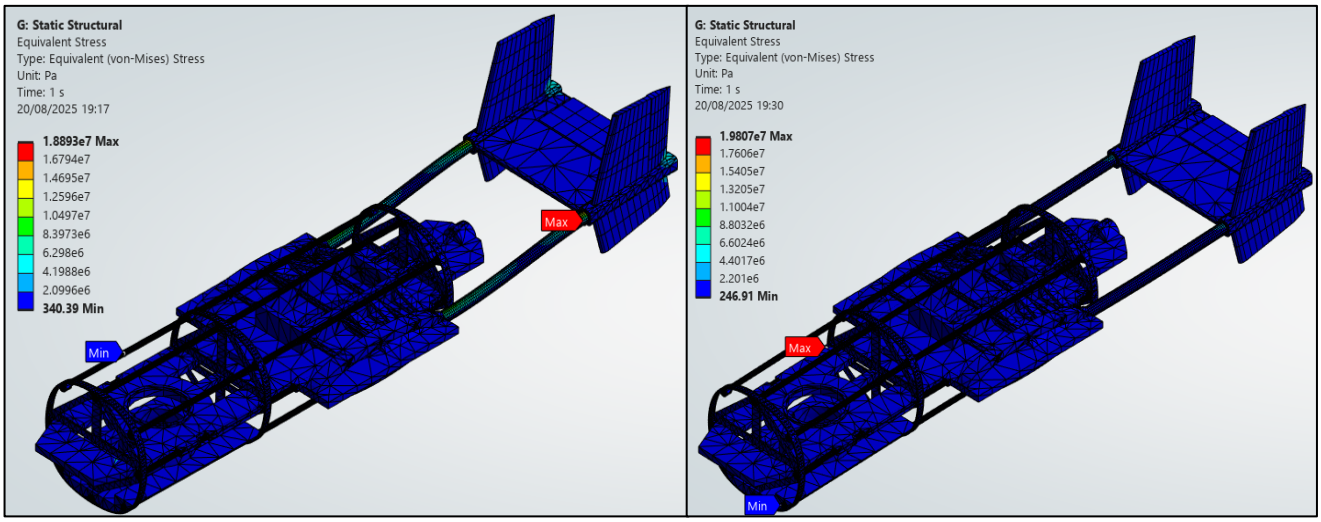


Figure 5.20: Equivalent Von-mises stress for Case 1(L) and Case 2(R) with a scale factor of 10

Strain Response

- $7.7\mu\epsilon$ (Case 1) and $55\mu\epsilon$ (Case 2)
- Strains were negligible, reflecting very high safety margins

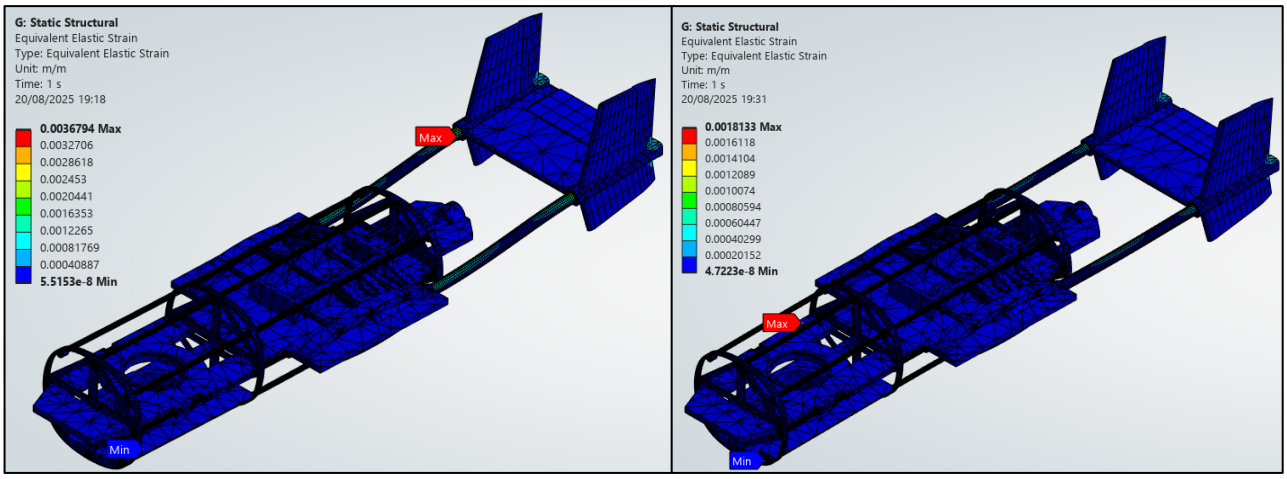


Figure 5.21: Equivalent strain for Case 1(L) and Case 2(R) with a scale factor of 10

5.7. Comparative Analysis

Table 5.2 below summarizes the comparative performances of all four structures.

Structure	Mass (kg)	Case 1 Deflection (mm)	Case 2 Deflection (mm)	Serviceability	Stress Safety Margin	Strain Safety Margin	Comments
A	14	0.97	4.95	Fail (Case 2)	High	Moderate	Too flexible under high load
B	12	0.12	0.97	Pass	Very High	High	Best balance of weight and stiffness
C	30	0.12	1.56	Fail (Case 2)	High	Low (local strain risk)	Needs ply-level reinforcement
D	35	0.02	0.16	Pass	High	Very High	Excellent stiffness, but heavy

Table 5.2

Discussion:

This comparison confirms that all four designs are structurally safe in terms of stress and strain, yet serviceability under high loads tends to be the discriminating factor.

- Structure A fails due to excessive flexibility large stress margins
- Structure B is a bit more efficient being able to maintain stiffness within limits while reducing weight.

- Structure C shows the trade-off of reinforcement: lower global deflection but creation of new localized strain hotspots which could lead to risk of ply damage.
- Structure D is the stiffest but there is an increase in weight due to the tail which could be excused. Nevertheless, it isn't practical in a lightweight design context

A critical point in this project was to establish the maximum and minimum dimensions of the autogyro. The selected rotor blades which were 1.2 m in length were very small compared to the size of the designed fuselage. This problem could have been addressed at the beginning or by scaling down the entire structure. The problem with scaling down the structure is that the cut-out for the payload would have had to stay the same which would then require redesigning the payload integration. A smaller structure could have offered solutions to the problems that came up while conducting FEA. The dominant theme here is geometry and stiffness drive viability but not material capacity.

5.8. Design Recommendations

Apart from scaling down the entire structure the results indicated that CFRP airframes are not limited by strength but by deflection and joint durability. Design improvements from here on must target:

- **Span Management:** Large-span members in Structures A and C require bracing or increased diameters to suppress bending.
- **Joint Reinforcement:** Tube-plate junctions and propeller mounts must be stiffened with gussets/doublers. This would help in reducing stress.
- **Laminate Tailoring:** Ply-level criteria should guide reinforcement to parts where the strain exceeds $1100\mu\epsilon$ for example in Structure C.
- **Mass Optimization:** Structure B's safety margins provide scope for ply-thickness reduction without loss in serviceability.
- **Dynamic Considerations:** Fatigue and modal analysis are required to assess cyclic and resonance behavior under rotor operation.

These measures transform the findings into pathways for optimized stiffness, durability, and weight balance

5.9. Alignment with Project Aims

The project aimed to deliver a lightweight, structurally efficient CFRP autogyro for medical

transport. The findings directly support and extend this goal:

- Strength capacity is not limiting: all designs are below CFRP's tensile and strain capacities.
- Deflection governs feasibility: serviceability limits define the critical performance threshold, reframing design focus toward stiffness optimization.
- Structure B with a shortened chassis emerged as the most promising solution in terms of just comparing the main chassis frames. It has a lower overall weight and remains functional even under load without excessive deformation.
- The study helps in contributing to design understanding by highlighting that geometry and laminate optimization techniques play a much bigger and important role than material capacity in dictating long-term reliability.

Thus, the results so far have confirmed that CFRP's suitability for medical transport applications while identifying stiffness control as the decisive factor to ensure its operational success.

6. CONCLUSION

The investigation was aimed to develop a lightweight, aerodynamically efficient autogyro airframe optimized specifically for rapid medical logistics within controlled urban environments. Through iterative design, simulation and prototype evaluation, the study has generated several important findings that inform structure optimization and future development.

Key Findings

Geometry-Stiffness Trade-off: The comparative FEA cases confirmed that structural geometry has a more decisive impact on serviceability over just increasing mass. The shortened chassis i.e. Structure B achieved the best balance, thus maintaining low (12kg) while limiting the deflections within serviceability limits under both load cases. By contrast, adding reinforcements as seen in Structure C reduces deflections only marginally and introduces strain risks. While in the heavy full-body design i.e. Structure D, although stiffer suffers from weight penalties.

Load-Case Sensitivity: The second load case simulated a worst-case asymmetric thrust and lift scenario which proved critical for exposing serviceability failures. Structures A and C exceeded the acceptable deflection limits under this case. This helped in emphasizing on the importance of evaluating rotorcraft airframes under extreme and non-uniform loading rather than solely relying on the nominal operating conditions.

Prototype Development and Practical Insights: Rapid prototyping and engagement with the stakeholders, who in this case were the NHS, accelerated design refinements. The best-performing configuration was Structure B which demonstrated deflections of 0.12mm under Case 1 and 0.97mm under Case 2. Both were within serviceability thresholds. High safety margins in stress and strain further confirmed its suitability for continued development.

Scaling Considerations: A critical issue identified was the mismatch between rotor diameter and fuselage dimensions. The selected motor offered limited lift capacity relative to the fuselage size. Addressing this like mentioned in the discussion section could be done through either scaling down the entire structure or redesigning the entire structure. Scaling down the structure could simplify the process, but it would mean redesigning the payload bay to maintain the required cut-out dimensions. Early integration of rotor sizing into the

overall geometry would prevent downstream redesigns.

Recommendations for Future Work

Building on these findings, some of the recommendations are:

- **Prototype Development:** Finalize on the shortened chassis (Structure B) as the baseline for full-scale prototype fabrication.
- **Laminate Tailoring:** Integrate ply-level reinforcement in high-strain regions which were identified in structure C to enhance the durability. This would also reduce any possible risk of localized failure.
- **Material and Sensor Validation:** Extend testing of self-healing laminates and embedded fibre-optic sensors under environmental conditions and fatigue. This would help in confirming long-term reliability.
- **Dynamic Testing:** Conduct a full-scale vibration, modal and aeroelastic stability analysis under rotor loads to understand how the body reacts and verify safe operations across the flight envelope.
- **Team Collaboration and Communication:** Implement structured design-review checkpoints and meetings within the team (e.g., CAD sessions for the payload, structures and power and lift team leads) to ensure critical information such as rotor sizing, and payload integration, and structural dimensions are aligned across all-team members.

Incorporating these technical and procedural improvements will only help in strengthening the autogyro's performance and reliability but also help build a cohesive development process ensuring that design decisions are evaluated and communicated smoothly from concept to full-scale deployment.

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8. PROJECT MANAGEMENT

Ref no	Title	Task Description	Due date	% completion
1	**Project initiation and planning**			
	Aims and Objectives Submission	Define project scope, goals, and key deliverables	W1 (13th March 25)	100%
	Literature Review	Analyze previous UAV airframe designs and structural challenges	W3	70%
	Requirement Analysis	Identify airframe constraints for integration with payload & lift subsystems	W4	100%
2	**Conceptual Design & Analysis**			
	Preliminary Airframe Design"	Develop a lightweight, aerodynamic structure to minimize drag	W6	100%
	Structural Load Analysis	Assess weight distribution and integration points for payload & lift	W7	100%
	Material Selection	Evaluate lightweight composites and alloys for structural efficiency	W8	100%
3	**Detailed Structural Design & Simulation**			
	CAD Modeling	Refine structural framework with stress-resistant features	W11	100%
	Structural Validation (Ansys FEA)	Analyze stress, vibration, and fatigue under operational loads	W12	100%
	Integration Considerations	Ensure airframe supports future payload and propulsion additions	W14	100%
4	**Validation & Testing**			
	Prototype Development	Fabricate a scaled airframe model for structural evaluation	W16	100%
	Structural Load Testing	Validate airframe integrity under simulated operational conditions	W18	100%
5	**Test Flight**			
	Assembly for Test Flight	Integrate airframe with temporary payload and propulsion for initial tests	W20	0%
	Test Flight & Structural Performance Analysis	Evaluate airframe stability, aerodynamics, and structural durability in real conditions	W22	0%
6	**Final Report & Presentation**			
	Report Writing	Document structural design, material selection, test results, and findings	W24(21st Aug 25)	100
	Final Presentation	Present airframe design, simulations, test flight insights, and integration consideration	W24	

9. SELF REVIEW

The autogyro medical delivery project did progress from the concept to a scaled prototype but over the course of the last 6 months there were strengths and weaknesses it highlighted.

I began this project with a clear schedule and was able to manage tasks effectively, which helped me meet many of the objectives I had set for myself. On the technical side I developed a structured FEA workflow to compare the four structural configurations, which strengthened my understanding design-tradeoffs. Working consistently on Fusion 360 also tested and improved my design skills, which I believe will help-me in the long run. Finally, building the prototype allowed me to see my CAD concepts turn into a physical model and gave me insights into the manufacturing etc.

With positives there were a lot of weaknesses exposed. First off was the weight constraint, Given, a maximum weight of 9kgs, the final model ended up more than 3 times heavier. This was a big failure in terms of not being design disciplined. Starting design process with the outer frame was another poor choice that I feel is what led to the structure being a lot heavier with a lot of dead space. Another major problem was the overall structure being too big relative to the rotor's diameter, even though it was mentioned well in the beginning. This could have been avoided early in the project if I communicated better. The payload system integration was slightly underdeveloped as there was no damping system to fully protect the payload which is the most important part of aircraft. Creating the iron bird was done towards the end when it should have been done earlier like mentioned in the methodology flow-chart.

What I take from this is that is that sticking to my constraints is vital. I should have built a simple prototype much earlier, even if it failed, to learn and improve. Over-engineering it was not the way to go.

Overall, the project gave me valuable insights and produced some results that the next teams can take forward when they work on this project. The experience has taught me to be more disciplined, to integrate different aspects earlier, and to plan testing around what really matters.